

Spanning tree

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Description:

Spanning tree is a tree structure by running dfs on a given graph. It has some interesting properties in the graph problem

Nature:

(1) Given a **connected component** with n vertex and m edges, the number of cycle is $m - n + 1$

Proof:

After generate the spanning tree of this component we define the edge on tree called **tree edge** and all other edge called **non-tree edge**, so obviously there is $m - n + 1$ non-tree edge and for any of them will form a cycle on the tree.

(2) Given a **connected component** with n vertex and m edges, the number of Eulerian subgraph is 2^{m-n+1} (inclusive empty graph)

Proof:

It can be show that for any Eulerian subgraph, it is a graph combine by any number of cycle, so according to the Nature 1 we know there is $m - n + 1$ cycle on the component and each of them either choose or not choose so the result is 2^{m-n+1}

(3) Given a **connected component** with n vertex and m edges, if a edge can be including in **minimum spanning tree** if and only if any cycle including it, it can't be the only maximum edge

Proof:

We will prove in two direction,

First, if the edge E is the only maximum edge in one of its cycle then it can't be including in the minimum spanning tree, the reason is simple if it been include in the MST then for those $m - n + 1$ non-tree edges there exist one edge E' such that include E' will form a cycle contain E but $val(E') < val(E)$ which mean if we give up E and choose E' in our spanning tree then we can get a more smaller value of MST which is contradiction.

Second, if the edge is maximum but not only maximum or it's not maximum then it can be include in MST, same logic we can find a edge E' in $m - n + 1$ non-tree edge such that it form a cycle contain E then if $val(E') = val(E)$ then no matter choose E' or E the value of MST won't change, if $val(E') < val(E)$ then obviously it's impossible